Marcelo Gonzales

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Summary

Mechanical engineer with 7 years of engineering experience in aerospace, embedded systems, hardware integration, and product development. Experienced in safety-critical aerospace environments, engineering analysis, hardware integration, troubleshooting, and verification. B.S. in Mechanical Engineering and Computer Science with hands-on experience designing test equipment, cable assemblies, and hardware-support systems.

Experience

The Boeing Company Houston, TX
Vehicle Management Software Engineer *May 2023 – July 2026*

CST-100 Starliner

- Developed and maintained embedded flight software for a human-rated spacecraft using C in a real-time operating system environment.
- Performed software integration, verification, debugging, and requirements-based testing for safety-critical spacecraft functions.
- Created and maintained test automation suite using Perl, Python and Jenkins.

MQ-25

- Developed and maintained C# test software for requirements-based testing and verification of MQ-25 mission-management flight software, supporting flight-readiness activities.

International Space Station

- Developed C and Python software tools supporting International Space Station flight operations and mission-support.
- Created and maintained database utilities used to manage, validate, and retrieve mission-support data.
- Legacy test frameworks were redesigned and refactored to increase test automation, improve requirements traceability, and strengthen software verification throughout the development life cycle.

Invocon, Inc. Conroe, TX
Software Design Engineer *October 2019 – March 2023*

- Developed embedded firmware in C using FreeRTOS for a smart-battery system.
- Developed and maintained C# and Python GUIs for real-time monitoring of hardware health and status.
- Built a Python library and developed standalone scripts to automate verification across multiple products, including SSI, KIPS, and iPCMe.
- Designed custom cable assemblies and test equipment to support hardware integration, functional testing, and product verification.

Education

Colorado School of Mines Golden, CO
B.S. in Mechanical Engineering *August 2019*
B.S. in Computer Science

Technical Skills

Mechanical: SolidWorks, 3D modeling, manufacturing drawings, BOMs, cable assemblies, test equipment, Mathcad, FEA.